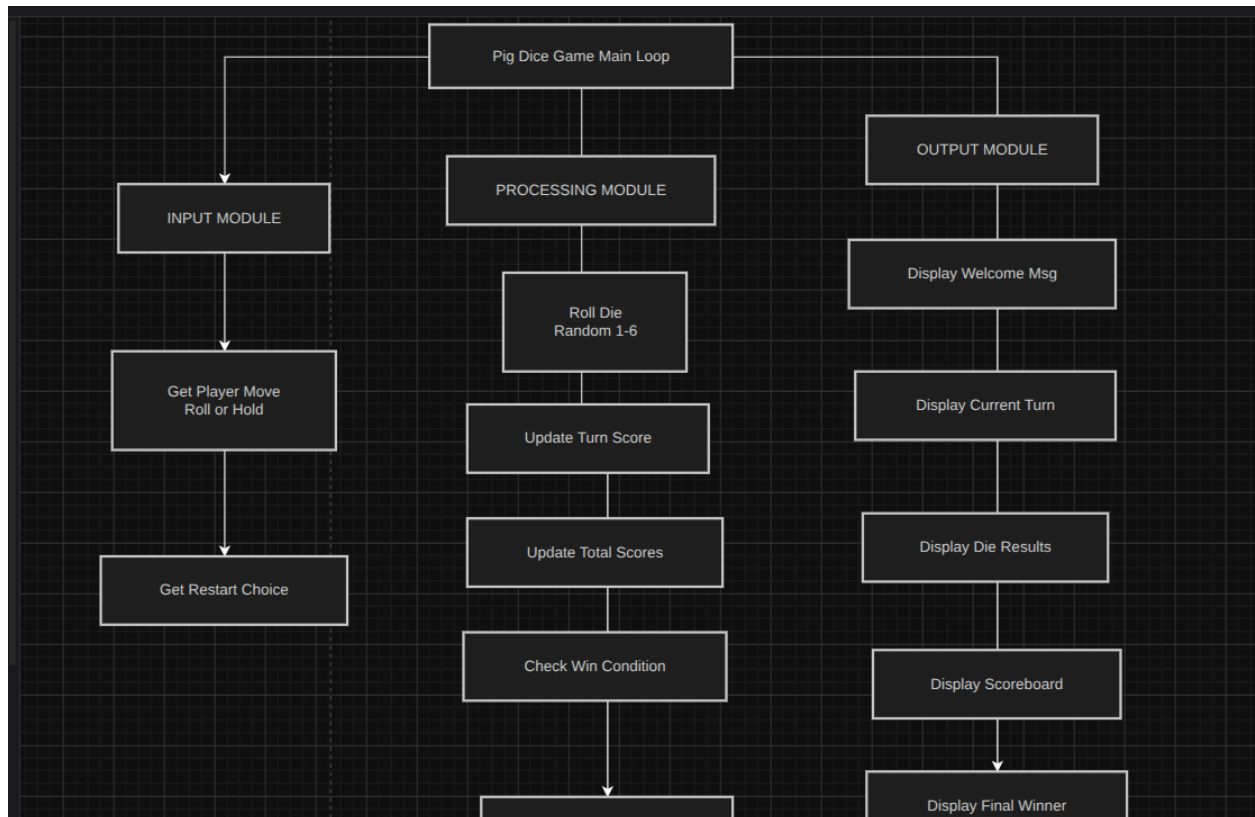


Pig Dice Game

The Pig Dice Game is a classic jeopardy dice game for two players where the ultimate goal is to be the first to reach a total score of 100 points. On each turn, a player repeatedly rolls a single six-sided die to build up a temporary "turn score," adding up the face value of any number rolled from 2 through 6. However, the game introduces a high-stakes risk element: if the player rolls a 1, their turn immediately ends, they lose all the points accumulated during that specific turn, and control passes to the opponent. To secure their hard-earned points, the player must strategically choose to "Hold" before rolling a 1, which permanently locks their current turn score into their permanent total scoreboard.

1. Structure Chart

The structure chart below delineates the modular breakdown of the system, separating tasks into standard Input, Processing, and Output components to ensure high maintainability and clean separation of concerns.

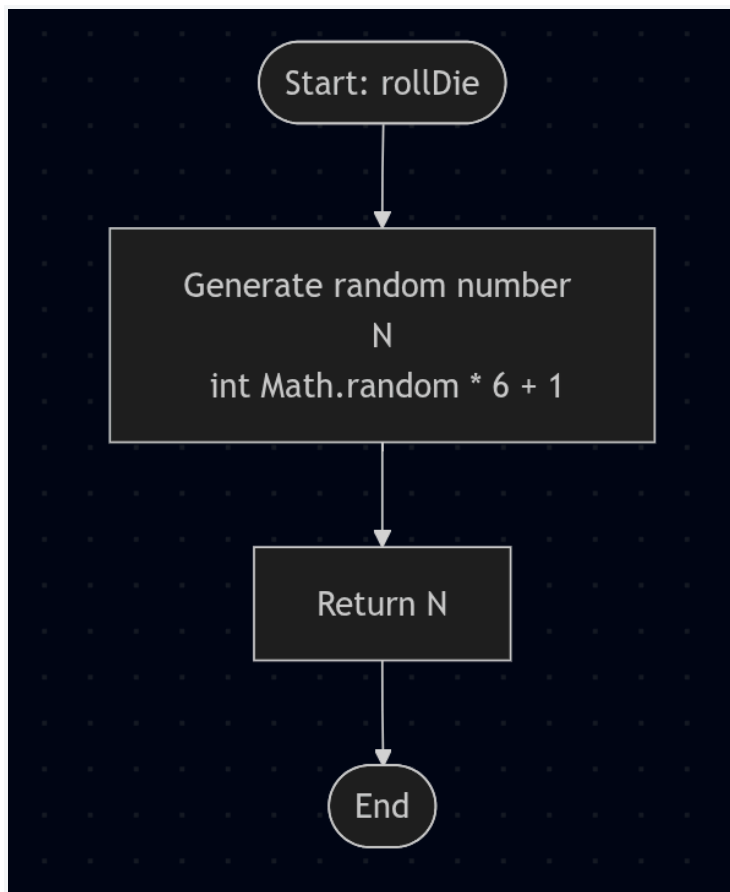


2. Program Flowcharts

The operational logic for the two primary programmatic subroutines governs how data transformations occur mid-game.

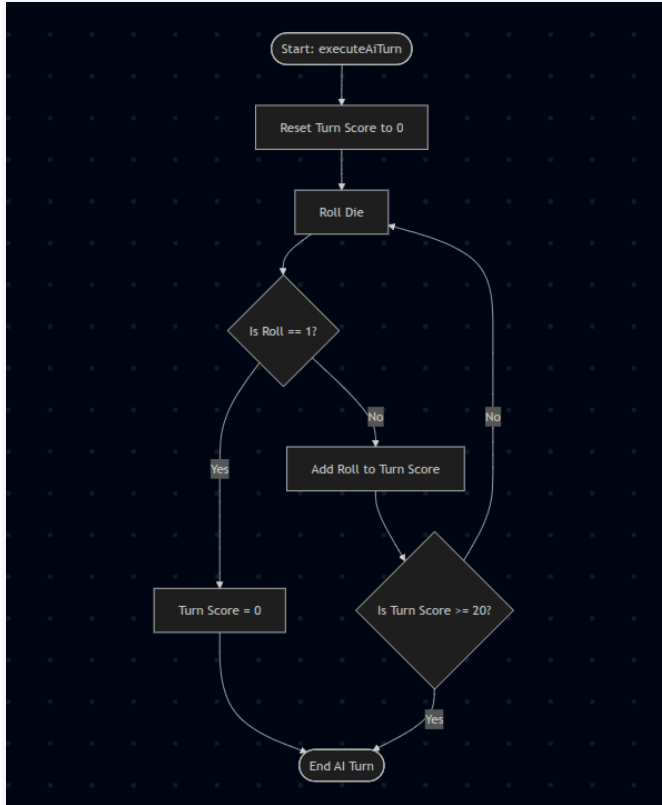
Subroutine A: rollDie() Logic

This module encapsulates the pseudo-random generation mechanics required to simulate a fair, six-sided die.



Subroutine B: `opponentTurn()` Logic

This subroutine manages the automated state machine dictating the behavior of the opponent based on predefined mathematical thresholds.



3. Software Quality Assurance Test Plan

The matrix below serves as the structural validation blueprint to ensure the runtime stability of the Java executable across edge cases and standard loops.

Test Case ID	Feature Tested	Input / Condition	Expected Functional Result	Status
Test 1	Die Boundaries	Call rollDie() 100 times iteratively	Every pseudo-random integer generated is strictly within [1, 6].	

Test Case ID	Feature Tested	Input / Condition	Expected Functional Result	Status
Test 2	Turn Logic (Roll 1)	Die sub-evaluation drops an active value of 1	Current turn score resets to 0; engine The opponent starts their turn.	
Test 3	Turn Logic (Hold)	Player requests "Hold" flag with active turn score at 15	Global scoreboard updates permanently (+15); engine yields control.	
Test 5	Win Condition Boundary	Player global tally increments to 102 points	Game-loop terminates breaks instantly; UI prints specific winner banners.	

4. Gantt Chart

Pig Dice Game 2-Day Project Timeline

